A STUDY OF 3,165 DOCUMENTS FROM EIGHT WORLD REGIONS · EXPANDED EDITION

Who governs the robots?

What the world’s laws, standards and research papers say about machines that act in the physical world, explained for people who are new to the subject, with the evidence behind every number

Centre for Enterprise Innovation (CEI), AI Robotics project
Version 4 of the study, 31 October 2026 · 41 slides

Collected, labelled and reviewed with AI agents; no person read every document

WHY THIS MATTERS

Why machines with bodies need their own look

Most writing about “AI rules” is about software: chatbots, recommendation systems, credit scoring. A robot, a drone or a self-driving car is AI with a body. When it goes wrong, the harm is physical, and the questions change.

Safety

Can it hurt someone? A 200 kg arm or a 1,500 kg car can.

Proximity

How close may it work to people: a factory cage, a hospital corridor, a sidewalk?

Liability

Who pays when it damages property or injures a person?

Autonomy

How much may the machine decide on its own, and when must a person step in?

The study calls this **embodied AI**: robots, drones, vehicles, surgical and factory machines. It asked what rules exist for them, anywhere in the world, and what those rules actually say.

Source: paper, Sections 1 and 2

WHAT CAME BEFORE

Two literatures that rarely met with numbers

AI ETHICS AND AI LAW

Reviews of ethics guidelines (84 in 2019, 36 in 2020, 200 by 2023) found the same handful of principles everywhere and little effect on practice. That pushed governments toward binding rules with risk tiers, testing before sale, audits and oversight bodies; the EU AI Act became the reference point.

All of it is about software.

ROBOT LAW AND ROBOT ETHICS

An older line of work argues that machines acting physically raise their own questions: bodily harm, closeness to people, liability for damage, how much autonomy to delegate, certification before market. Its reference points are the RoboLaw guidelines (2014), the European Parliament's 2017 resolution, the ISO robot-safety standards and Korea's 2008 Intelligent Robots Act.

Mostly expert argument, rarely counted.

What existing databases do not do

The OECD.AI observatory lists over a thousand policies but does not separate machines with bodies or code what each text says; legislation trackers record events, not content; the robot-law mappings are hand-coded and European. This study is the first coded collection that separates embodied from general-AI governance, scores six body-specific concerns, and reports how its own counts move with the counting unit and the search strategy.

Source: paper, Section 2 (Background)

THE STUDY

The questions we asked

No.	In plain words	Why it is hard
1	What rules for robots exist, and how many?	The answer changes with what you decide to count as “one rule”.
2	Which robots are governed, and which are only studied?	Researchers and lawmakers pay attention to different machines.
3	What makes a robot rule different from a general AI rule?	Robot rules and AI rules come from different kinds of documents.
4	Do regions of the world differ?	Each region’s documents are a different mix of papers, laws and standards.
5	What is changing over time?	Old laws are amended; dates can mislead.
6	Does how you search change what you find?	This question arose from the study itself. The answer is yes, a lot.

Source: paper, Section 1

HOW IT WAS DONE

How the documents were collected

ROUND 1: PEOPLE SEARCHING THE WEB

Eight regional teams (Africa, Asia, Europe, Latin America, Middle East, North America, Oceania, international bodies) searched for anything about robot and AI governance.

They found research papers, strategies and guidelines, but few actual laws.

1,656

documents after duplicates were removed

ROUND 2: AI AGENTS SEARCHING LEGAL DATABASES

Nine AI search agents and seven scripts queried official law databases (EU, UK, Japan, China, Australia, US, Brazil), standards catalogues and treaty archives for the laws the first round missed.

2,075 candidates, 1,720 texts obtained, 1,626 usable.

1,641

documents added to the collection

TOGETHER

3,295

documents, 3,165 with enough readable text to use

116

countries, plus the EU and international standards bodies

75% in English; Chinese is the second language

Source: paper, Section 3; round-2 companion report, Figure 50

HOW IT WAS DONE · ROUND 1

Round 1: eight teams searching the web

Between 2025 and October 2026, regional teams for Africa, Asia, Europe, Latin America and the Caribbean, the Middle East, North America and Oceania, plus an international layer assembled across teams, ran two exercises.

The sprint

Robot-specific sources: anything about robots, drones, vehicles and their governance.

The database

General AI-governance instruments: strategies, guidelines, laws about AI in general, as the comparison group.

The funnel



266 sources could not be captured, 109 of them in the European sprint. Duplicates were found by content hash, URL, normalised title and text similarity; 27 of 34 oversized or restricted files were recovered.

What the teams found: research papers, strategies and guidelines. What they missed: the aviation, vehicle and machinery registers where binding law lives. Round 1 held 65 robot instruments, 5 of them binding.

Source: paper, Section 3 (Collection round 1); Table 53

HOW IT WAS DONE · ROUND 2

Round 2: AI agents and scripts searching legal databases

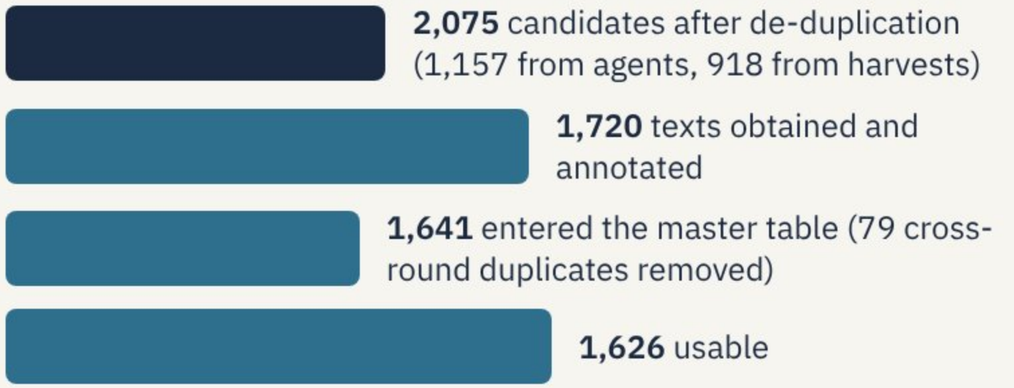
Prompted by the first round's coverage audit, a second round looked where the law lives. Nine AI search agents, each with a regional, thematic or standards brief and the round-1 holdings as an exclusion list, plus seven scripted harvests of official databases. No person curated the candidates.

The seven databases

EUR-Lex (EU law) · legislation.gov.uk · Japan's e-Gov law API · China's State Council policy library · Australia's Federal Register of Legislation · the US Code of Federal Regulations · Brazil's Chamber of Deputies API

355 candidates never yielded a text: 272 returned errors or script-only pages, 46 sat on hosts that block automated access, 37 failed otherwise. They are counted nowhere.

The funnel



Of the 1,720 texts, 1,704 came from the issuing source or a fallback route and 16 standards' scope pages through a mirror. Every text was labelled with the same metadata-blind, long-excerpt policy as the first round's governance documents.

Source: paper, Section 3 (Collection round 2); round-2 companion report, Tables 50 to 52 and Figure 50

HOW IT WAS DONE

How the documents were labelled

An AI model (Claude) read every document against a fixed questionnaire of 39 fields, the same for all 3,295 documents: What kind of document is this? Is it legally binding? Which machines does it cover? Which safety and ethics topics does it mention, and which does it spell out in detail?

The model saw only the text, not the collector's notes, so it could not be steered by a label someone had already given.

No person read or labelled any document. The collectors in round 2, the reviewers of earlier drafts and the auditor were also AI agents. Treat every label as consistent, not certified.

How consistent are the labels?

Agreement is measured with kappa: 0 means two readers agree no more than chance, 1 means they always agree. Above 0.8 is usually called strong.

Check (160 documents each)	Round 1	Round 2
A second, different model	0.85	0.86
The same model, run again	0.91	0.91

Weakest fields: judging page quality, a document's stance toward regulation, and whether a law is still in force.

Source: paper, Sections 4 and 5.8; supplement S4.7

HOW IT WAS DONE

The 39 questions asked of every document

Who wrote it and when (10)

- Title, cleaned
- Title in English
- Language of the text
- Author or issuing body
- Kind of issuer (16 kinds)
- Year of publication
- Month of publication
- Where the date comes from (3 options)
- DOI, for research papers
- Journal or venue

Stance and method (3)

- Stance toward regulation (5 options)
- Research method, if a study (9 kinds)
- Faith or religious perspective (yes or no)

Where it applies (3)

- Country or body it belongs to
- Scope: local, national, regional, global (5)
- Other countries it discusses

What kind of document (6)

- Document type (28 types)
- Document class (8 classes)
- Is it a governance document? (yes or no)
- Status: proposed, in force, repealed … (11 states)
- Binding nature (5 options)
- Publication stage (7 stages)

Quality of the reading (5)

- Page quality: full text, excerpt, index page … (5)
- Relevance to embodied AI (0 to 3)
- Summary in English
- Key provisions or claims
- The model's own confidence (0 to 1)

AI and machines (5)

- AI scope: general AI, robots, not AI … (5)
- Embodiment status (4 options)
- Robot types mentioned (23 types in 7 groups)
- Main robot type
- Sectors of application (16 sectors)

What it says (7)

- Governance topics (27 topics)
- Main topic
- 20 principles, each scored absent, mentioned or spelled out (14 ethical values, 6 about the body)
- Governance tools used (19 tools)
- Other laws or standards it cites
- Position on legal personhood for robots (4)
- How it frames risk (7 framings)

Source: codebook and annotation schema (supplement S2); every field has a fixed list of allowed answers where a count is given

HOW IT WAS DONE · A REPAIR

The Chinese, Japanese and Korean texts had to be read again

Before a document reaches the model, a cleaner strips menu lines and navigation from the top of web pages. The rule used in the earlier version dropped short lines that had no spaces between words and no Western full stop. Chinese, Japanese and Korean paragraphs have neither.

So for most Chinese documents only a fraction of the text reached the model. A reviewer measured it: a median of a quarter of each Chinese document was read, and 83 Chinese documents that name robots or AI were labelled “not about AI”.

The fix: a character-aware cleaner that never drops a line carrying Chinese, Japanese or Korean characters, and a re-reading of every such document under the same policy.

422

documents re-read: 376 from round 2, 46 from round 1

57

changed whether they count as about robots or embodied AI

205 → 160

documents rated “not about AI” before and after the repair

Source: paper, Section 3 (Processing and corrections); supplement S4.7; cost of the repair 2.5 USD

HOW IT WAS DONE · CORRECTIONS

Four rules and 157 corrections

An expert reviewer read 159 risk-stratified entries of the earlier version and found at least one error in 68% and a material one in 62%. The sample was chosen where errors were likely, so those rates are a ceiling, not the average. The response was in two parts.

Four rules, applied to every document

- A standard still in development (draft stages) is not a published instrument.
- A mandatory Chinese national standard (GB, not GB/T) is binding law.
- A working paper or chair’s text from the UN weapons talks is a position document, not a draft rule.
- A proposal whose adopted act is in the collection is superseded, not “under consideration”.

157 named corrections on 91 documents

Each one records the field, the new value, the reason and the source: 50 on the counting unit, 32 on relevance, 32 on robot type, 17 on status, 14 on class, 7 on year, 5 on binding nature. They live in a versioned file applied last, so anyone can see what was changed and why.

One example: the 2006 Machinery Directive had been marked repealed. It is in force until January 2027, and was corrected.

Source: paper, Section 3 (Processing and corrections); supplement S8; data/manual_overrides_v4.csv

BEFORE THE NUMBERS

Five words you need

Word	What it means here	Example
Instrument	Any official text meant to steer conduct, from a statute to a voluntary guideline.	A drone regulation; a national AI strategy; a safety standard.
Binding	Has legal force. You can be fined or stopped for ignoring it.	The EU Machinery Regulation; Japan's Road Traffic Act.
Soft law	Guidance, strategies and standards. Followed by choice or by contract, not by law.	The OECD AI Principles; ISO 10218 for industrial robots.
Principal instrument	A law or regulation counted once, with its amendments, without its paperwork.	Paperwork not counted: commencement orders, fee rules, exemptions, corrigenda, index pages, stubs under 150 words.
Embodied set	The documents the model judged to be about AI in a body, or tagged with a robot type.	1,603 of the 3,165 usable documents; 505 of them are principal instruments.

Every number on the following slides says which of these it is counting. That turns out to matter more than anything else in the study.

Source: paper, Section 3 (counting unit, construct and capture tier)

THE COLLECTION

What the two sets are made of

The model sorted the 3,165 usable documents into a robot set, a general-AI set and the rest. The two sets are built from different kinds of documents, which is why every comparison later is also made within one kind at a time.

ROBOT SET · 1,603 DOCUMENTS

Kind of document	Count	Share
Research papers	509	32%
Principal instruments	505	32%
Paperwork, index pages and stubs of instrument class	305	19%
News, reports, grey literature	284	18%

Issuers: academics 516, national executives 276, standards bodies 228, intergovernmental bodies 217, regulators 179, legislatures 91.

GENERAL-AI SET · 1,015 DOCUMENTS

Kind of document	Count	Share
News and communications	380	37%
Soft-law instruments (strategies, guidelines)	248	24%
Principal instruments of any kind	372	37%
Research papers	48	5%

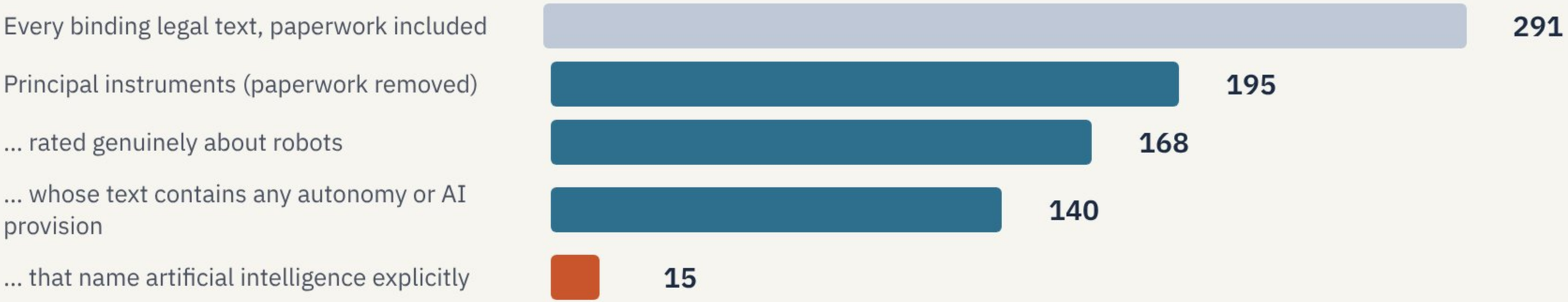
547 further documents are in neither set: the model judged them not to be about AI at all, 478 of them legal texts such as harvested statutes with no AI content.

Source: paper, Section 3 (Analysis sets) and Section 5.3, Figure 1. Principal instruments in the general set overlap with the soft-law row

COUNTING

How many rules there are depends on how you count

One collection, read under successively stricter definitions. Each bar is a count of binding texts in the embodied set; the bar length is proportional to the count.



The 96 texts that leave between the first two rows are commencement orders, levies, exemptions, index pages, duplicates and stubs: the way legal databases expose one regime as many entries.

Source: paper, Table 53b

THE NUMBER TO REMEMBER

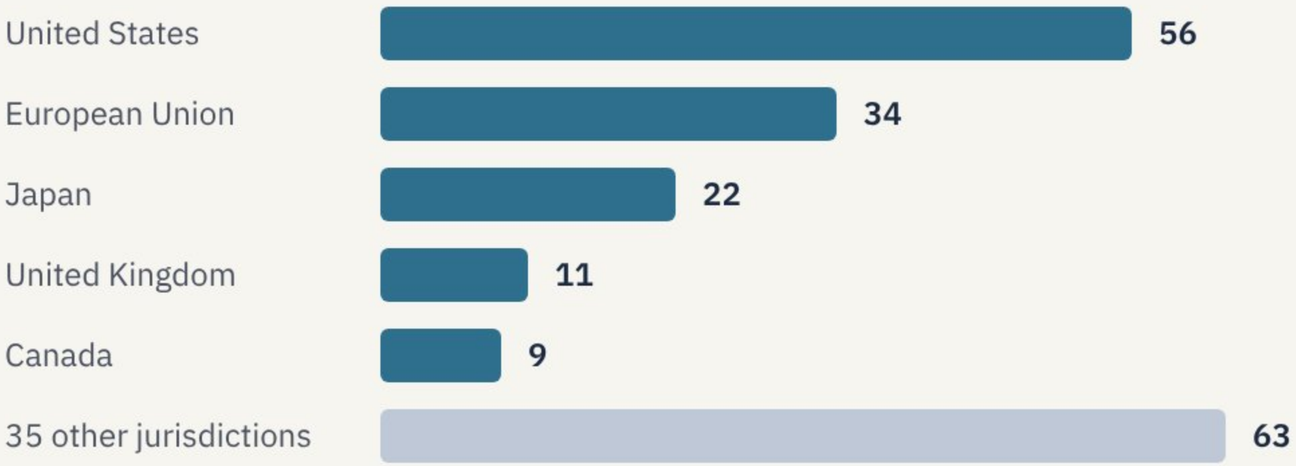
15 of the 195 binding rules that reach robots name artificial intelligence at all

The other 180 govern drones, vehicles and machines as machines. They were written for aviation, road traffic, factories and products, and reach robots because robots are also aircraft, vehicles and products.

THE 195 BINDING RULES

Where the binding rules come from

By jurisdiction



The collection over-represents places with open legal databases. France, Korea, Taiwan, Hong Kong and the Chinese provinces are thin because their law portals blocked the search.

By kind

132 regulations or decrees
60 statutes passed by a legislature
3 court decisions

By status

182 in force
9 superseded or repealed
4 adopted but not yet in force

Where they were found

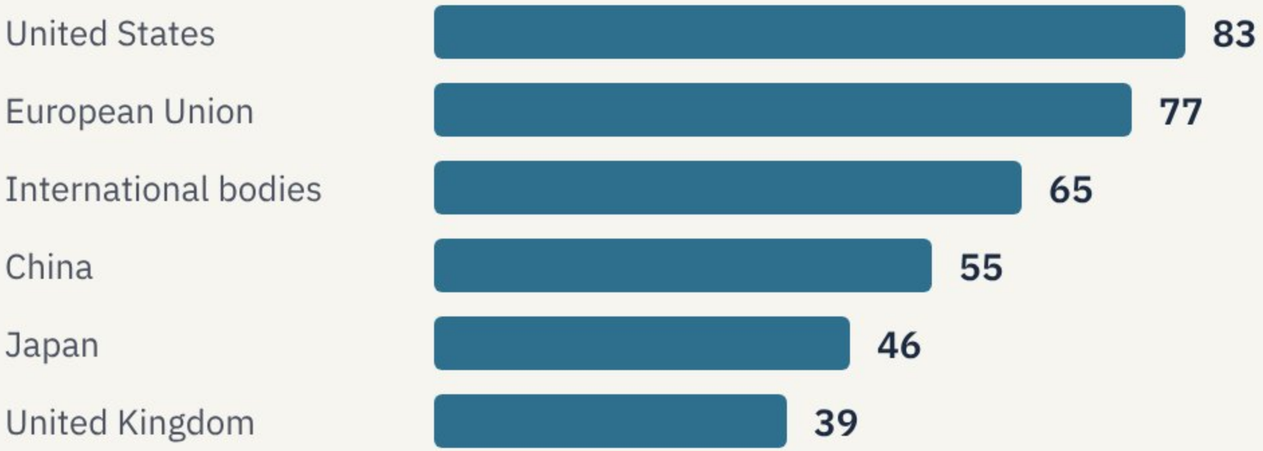
190 of the 195 came from the second, database-driven round. The first round, searching the web, had found 5.

Source: paper, Section 5.3 and Table 33; Section 7

THE COLLECTION

Which jurisdictions hold the robot rules

Principal robot instruments (binding, soft and draft) by the jurisdiction that issued them. Of 116 countries with a document, 58 hold a principal robot instrument and 40 hold a binding one.



International bodies means the standards organisations and treaty bodies; their 65 are mostly technical standards read from scope pages.

42%

of all usable documents are European (1,332 of 3,165); half of those are about robots.

68%

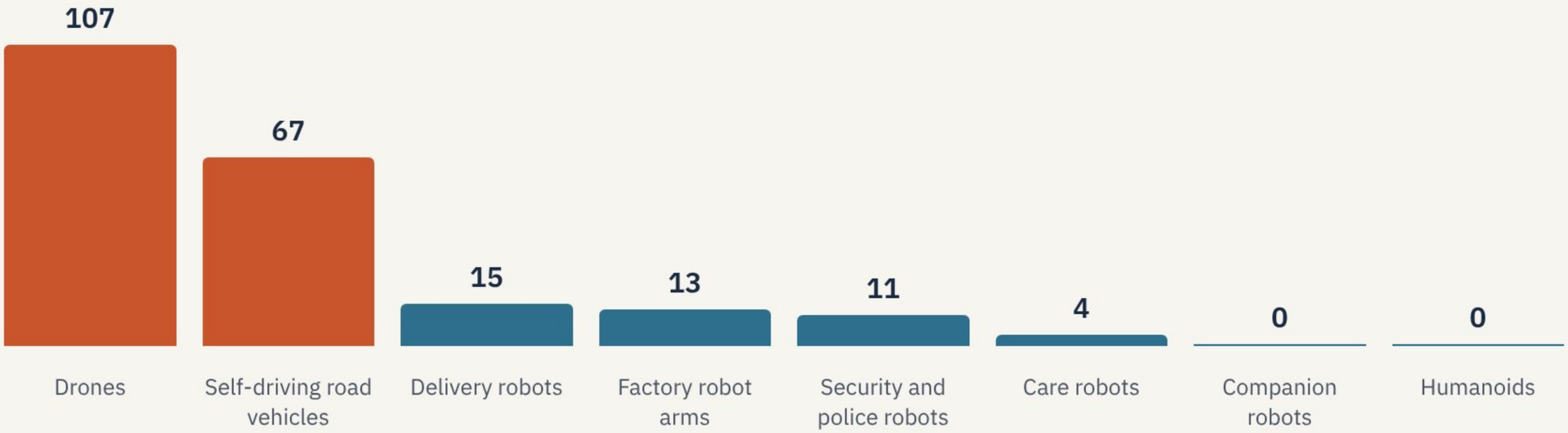
of the 792 non-English documents are governance documents, against 49% of the English ones. The law came in other languages; the research came in English.

Source: paper, Section 5.3 and Figure 7; Table 4

THE 195 BINDING RULES

Which machines the binding rules cover

Number of the 195 principal binding instruments that mention each kind of machine. One rule can mention several kinds, so the bars do not add up to 195.



Source: paper, Section 5.3 and 5.4; brief. Mobile logistics, agricultural, collaborative and surgical robots and autonomous weapons: 9 or fewer each

THE ROUTE

Law reaches robots sideways

Almost no country has written a law about robots. Instead, robots are caught by laws written for the things they also are: aircraft, vehicles, machinery, products, medical devices.

Aviation law

Drone rules: registration, pilot licences, no-fly zones.

Road-traffic law

Automated-driving amendments: who is the driver when nobody drives.

Machinery and workplace safety

Factory arms and collaborative robots as machines with CE marks.

Product and medical-device law

Surgical robots and consumer robots as products that must be safe.

General AI law

The EU AI Act reaches robots through its product lists, not by naming them.

The one robotics statute the study has read

El Salvador’s Robotic Technologies Law (2025), the only binding rule in the collection with “robot” in its title whose text was obtained.

The ones it could not read

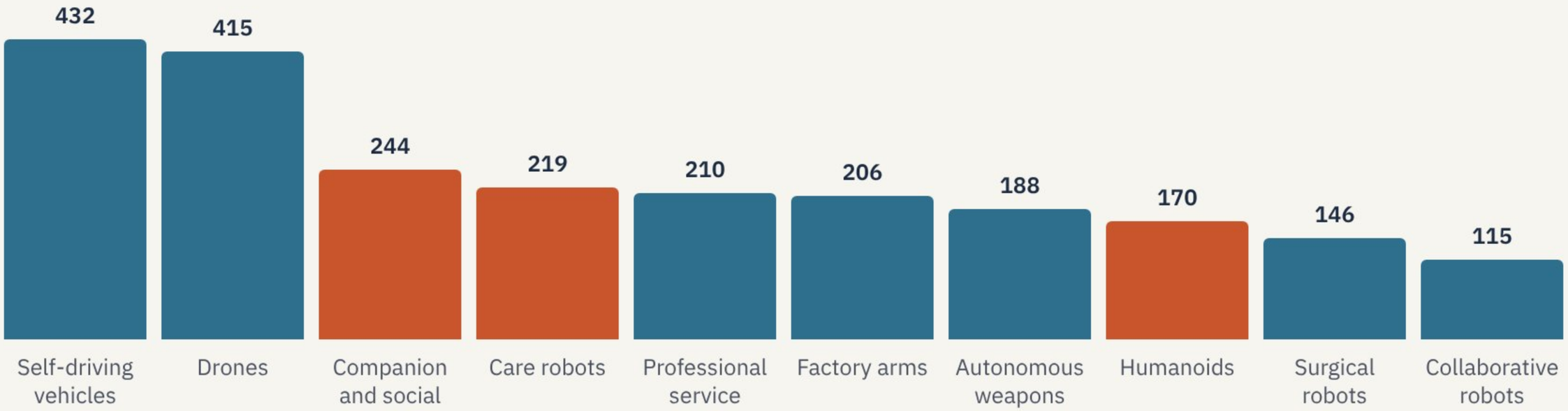
Hangzhou’s 2026 embodied-robot regulations and Korea’s Intelligent Robots Act (2008) were confirmed to exist but sit on websites the search could not reach.

Source: paper, Sections 5.3 and 6

WHICH ROBOTS

Which robots the documents talk about at all

Number of the 1,603 robot-set documents (papers, rules, reports and news together) that mention each kind of machine. A document can mention several.



Orange bars: the robots people will live with (companion, care, humanoid), which the next slides show are the ones the law has least to say about.

Source: paper, Section 5.4 and Figure 11

STUDIED OR GOVERNED

Research and rules point at different robots

Share of 509 research papers and of 505 principal instruments that address each kind of robot. The last column is the difference in percentage points, with the range it could plausibly take.

Kind of robot	Research papers	Instruments	Binding	Gap (points)
Companion and social robots	43%	3%	0	**+40 (36 to 45)**
Humanoids	23%	4%	0	**+19 (15 to 23)**
Care and assistive robots	28%	9%	4	**+19 (14 to 23)**
Factory robot arms	near parity	near parity	13	−3 (−7 to 1)
Self-driving road vehicles	12%	40%	67	**−27 (−33 to −22)**
Drones	6%	42%	107	**−36 (−41 to −31)**

Scientists study the robots people will live with; lawmakers regulate the things that move through public space. Part of the drone and vehicle lead is the search itself, which looked for those terms in legal databases.

Source: paper, Section 5.4, Table 22 and Figure 40

WHICH ROBOTS

Different regions, different robots, and a shift over time

Where each robot is talked about most

Kind of robot	Share of the region's robot documents
Companion and social	Europe 27% · North America 15%
Humanoids	Middle East 22% · Asia 15%
Drones	Oceania 65% · Africa 63% · Latin America 53%

Drones dominate the regions whose documents are mostly aviation regulation; social robots belong to the regions that produce the research.

Within the robot set, 2020 to 2022 against 2025 to 2026

FALLING

20% → 12%

documents that mention companion and social robots

RISING

8% → 17%

documents that mention humanoids

Attention is moving from the robot as companion to the robot as general-purpose worker. The humanoid gap in binding rules (none) may matter more each year.

Source: paper, Section 5.4; Figure 42 in the analysis report

ROBOT RULES VERSUS AI RULES

What makes a robot rule different

Share of principal instruments that raise each concern: 505 robot instruments against 372 general-AI instruments. The body is what sets robot rules apart.

Concern	Robot instruments	General-AI instruments	In plain words
Physical safety	**83%**	9%	Can it hurt someone?
Conformity assessment	**66%**	19%	Must it be tested before sale?
Autonomy level and control	**60%**	10%	How much may it decide alone?
Human-robot proximity	**33%**	0.3%	How close may it work to people?
Liability for physical damage	**27%**	3%	Who pays for the damage?
Dual use and military	**21%**	12%	Could it become a weapon?

These six differences hold in both collection rounds, in research papers as well as rules, and after statistical adjustment. The study calls them the most reliable result it has.

Source: paper, Section 5.5, Table 21 and Figure 39

ROBOT RULES VERSUS AI RULES · THE STATISTICS

Does the difference survive when like is compared with like?

A statistical model compares robot documents with general-AI documents of the same kind, region, year and collection round, across 2,526 documents. The odds ratio says how many times more likely a concern is to appear in a robot document once those differences are held constant. One means no difference.

Far more likely in robot documents

Concern	Odds ratio
Physical safety	**45**
Liability for physical damage	**21**
Autonomy level and control	**17**
Conformity assessment	**6.0**
Dual use and military	**3.3**
Non-maleficence (avoiding harm)	**2.6**
Human oversight	**1.7**

Less likely, and how to read that

Eight core values are less likely in robot documents after adjustment, from justice and fairness (0.27) to sustainability (0.65). Four show no difference: accountability, freedom, dignity, robustness.

Without the collection round in the model, eleven looked lower. The three that the round removes (accountability, freedom, dignity) are exactly the ones the within-round check showed to be artefacts of the second round.

Read the remaining eight as a property of harvested legal texts, not of robot governance as such. Proximity to people is not listed: its ratio rests on four general documents and is flagged as not estimable.

Source: paper, Section 5.5, Table 20 and Figure 38; Table 34 for the comparison without the round

ROBOT RULES VERSUS AI RULES

The tools robot rules use

Robot rules borrow the toolkit of product safety. General-AI rules reach for institutions and transparency. Share of principal instruments using each tool.

Used more by robot rules

Tool	Robot	General AI
Reference to a technical standard	**65%**	35%
Licensing and registration	**44%**	10%
Testing before sale (conformity)	**42%**	12%
Compulsory insurance	**15%**	2%

Think of a car: a type approval, a registration plate, an insurance policy.

Used more by general-AI rules

Tool	Robot	General AI
Creating an oversight body	lower	**higher**
Audits and impact assessments	lower	**higher**
Labelling and disclosure duties	lower	**higher**
Voluntary codes, funding, education	lower	**higher**

Think of a data regulator: an agency, an audit, a notice that you are talking to a machine.

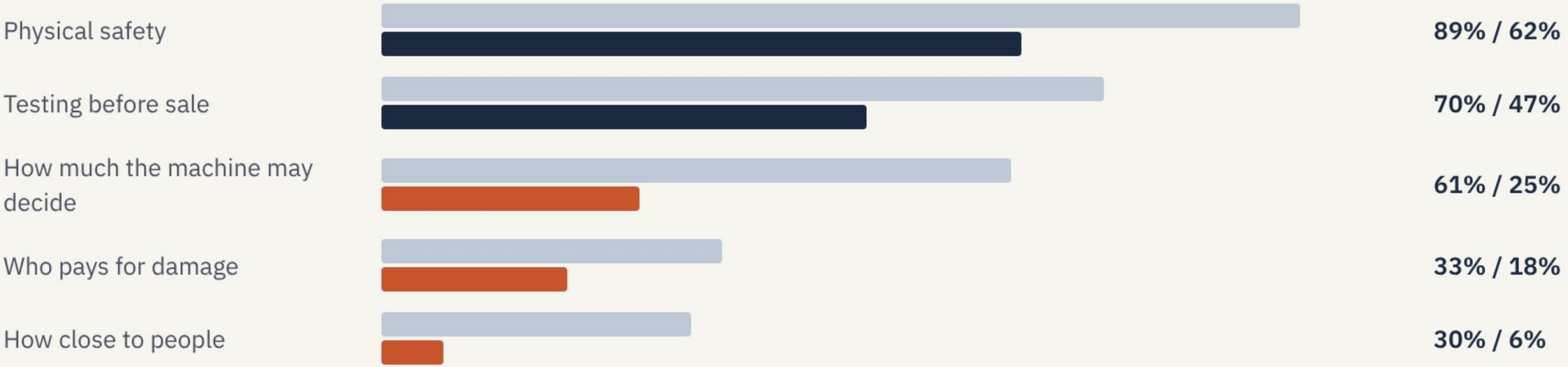
Risk tiers, sandboxes and human-oversight duties are used about equally by both.

Source: paper, Section 5.5, Table 21 and Figure 24; all differences significant after correction

MENTION VERSUS DETAIL

Mentioning a problem is not the same as solving it

For each concern, the share of the 195 binding robot rules that mention it (light bar) and the share that spell out what must be done (dark bar).



Binding law is good at safety and pre-market testing. The control question, how much a machine may decide and when a person must step in, is raised by six rules in ten and answered by one in four.

Source: paper, Section 5.5 and Table 9, content-analysed captures only

A SEPARATE MEASUREMENT

Technical standards were read from the cover, not the contents

Standards such as ISO 10218 (industrial robots) or ISO 13482 (personal care robots) are the operative safety texts for robots, but their full text is sold, not published. The study could read only each standard's public scope statement or abstract, often a few hundred words.

So every statement about standards describes what a scope page says, never what a normative clause requires, and standards are reported as their own measurement, never pooled with statutes.

Of the 810 robot legal texts, 222 are catalogue or landing pages, mostly standards listings. They enter no instrument count and no depth statistic.

What 62 standards' scope pages say

Concern	Mentioned	In detail
Physical safety	95%	68%
Autonomy level and control	60%	23%
Liability for damage	3%	0%
Dual use and military	0%	0%

For comparison, the 195 binding rules specify physical safety in 62% and autonomy control in 25%; soft-law texts other than standards (207 captures) specify physical safety in 33% and autonomy in 21%.

Source: paper, Sections 4 and 5.5, Table 9 and Figure 23

WHAT THE DOCUMENTS POINT TO

The two worlds cite different rulebooks

Share of documents in each set that name a given instrument. General-AI documents cite a shared canon; robot documents cite it far less and reach for different texts.

Instrument named	Robot documents	General-AI documents
EU AI Act	6%	**28%**
GDPR (EU data-protection law)	8%	**21%**
OECD AI Principles	lower	**14%**
UN weapons talks (CCW process)	**6%**	0.2%
European Parliament’s Civil Law Rules on Robotics	**4%**	0.6%
Any named instrument at all	27%	**52%**

Half of general-AI documents anchor themselves to a named rule; three robot documents in four name none. The robot field has sector law, standards and research, but no shared reference text.

Source: paper, Section 5.5 and Figure 29

HOW THE ISSUE IS FRAMED

Ethics in Europe, risk in North America, liability in the Middle East

The dominant frame, by region

Among robot documents that take a frame at all, the share using each one.

Region	Leading frame	Share
Europe (546)	Ethics-led	36%
North America (164)	Risk-led	31%
Oceania (53)	Risk-led	64%
Asia (260)	Innovation 38% · risk 34%	split
Middle East (41)	Liability-led	59%

What researchers say about regulation

Of 480 robot research papers that take a position:

56% take no regulatory position

31% call for new regulation

4% argue existing law suffices

The research community rarely engages with the law that already applies: only one paper in twenty-five argues that it does. The gap for companion robots is partly a gap in that engagement.

Source: paper, Sections 5.5 and 6

THE METHOD LESSON

Where you look changes what you find

VERSION 2: WEB SEARCH

5

binding robot rules found. Wrong: the laws live in legal registers, not on the open web.

VERSION 3: LEGAL DATABASES

299

binding texts counted. Inflated: every fee rule and exemption counted as a law.

VERSION 4: COUNTED PROPERLY

195

principal binding instruments. Neither jump happened in the world; both happened in the method.

The search also changed a conclusion

Version 3 reported that robot rules talk less about fairness, transparency and privacy than general-AI rules do, in 11 of 14 core values. Looked at one round at a time, the difference appears only in the second round: among the first round's instruments, none of the 14 values differs; among the second round's, 10 are lower. Harvested legal texts are short and procedural; strategies found by web search are long on values. The "ethical deficit" was a property of the documents, not of robot governance.

Source: paper, Sections 5.2 and 6, Tables 53 and 20b

AROUND THE WORLD

How regions differ

On tools: clearly

Whether a document refers to technical standards, requires a licence or sets a liability regime varies most across regions. Oceania's documents are drone-licensing law; the international layer is standards; Europe mixes research, strategies and regulation.

On harm: hardly at all

Physical safety, avoiding harm and human oversight are mentioned at similar rates everywhere. The harm-avoidance vocabulary travels.

On values: it depends what you compare

Among research papers, regions agree on fairness, transparency, privacy and beneficence (2 of 14 core values differ). Among instruments, 12 of 14 differ, with the standards-heavy international layer at the low end and Asia's strategies at the high end. So the difference is about what kind of document each region produces.

Which region holds the documents

Region	Usable docs	Robot rules
Europe	1,332	170
Asia	677	119
International bodies	374	64
North America	345	102
Latin America	149	21
Oceania	113	11
Middle East	98	8
Africa	76	10

"Robot rules" are principal instruments in the embodied set. Europe is 42% of everything; the collection is not a census of the world.

Source: paper, Sections 5.3 and 5.6, Tables 29 and 29b

OVER TIME

What is changing

RISING

+31%

a year in the odds that a document uses **risk tiers** (stricter rules for riskier machines), within the first collection; pre-market testing rises 26% a year. Both diluted, not reversed, by the harvested legal texts.

SHIFTING

0.39

odds ratio for the **robot personhood** debate in 2025 to 2026 against 2020 to 2024: it is fading. **Industrial policy** (subsidies, national champions) rises instead, odds ratio 1.69.

ARRIVING

13%

of 2026 documents about robots use the terms **“embodied AI”** or **“physical AI”**, up from 1% in 2024 and 8% in 2025. The vocabulary is new; the machines are not.

A trend that turned out not to be one

Pooled over both rounds, attention to “how close to people” and “who pays for damage” looked like it was declining 6% a year. Within the first collection it is flat. The decline was the arrival of short legal texts that never mention either, not a change in what people write about.

Source: paper, Section 5.7, Tables 28 and 30, Figure 33 in the report

THE EVIDENCE · CONSISTENCY

Seven ways the labels were checked against themselves

Kappa measures agreement beyond chance: 0 is chance, 1 is perfect, above 0.8 is usually called strong. Each check re-labels a sample of 160 documents a different way and compares.

Median agreement with the final labels

Check	Kappa
Round 1: a second, different model	0.85
Round 1: the same model run again	0.91
Round 1: the same model reading the full text	0.88
Round 1: with all metadata hidden (leakage check)	0.90
Round 1: the longer re-annotation of governance texts	0.83
Round 2: a second, different model	0.86
Round 2: the same model run again	0.91

Where agreement is weakest

Judging page quality (0.53), a document's stance toward regulation (0.65), its publication stage (0.66), where its date comes from (0.70) and whether a rule is still in force (0.75). Status and class define the analysis sets, and they agree less well in round 2 than in round 1.

Two further checks

The round-2 collectors' own catalogue data agree with the blind labels on binding nature in 93% of 1,608 cases, on jurisdiction in 98%, on year in 94% and on document class in 77%. Simple word lists over the full text agree with the labels at a kappa of about 0.4, but rank the principles the same way (correlation 0.88).

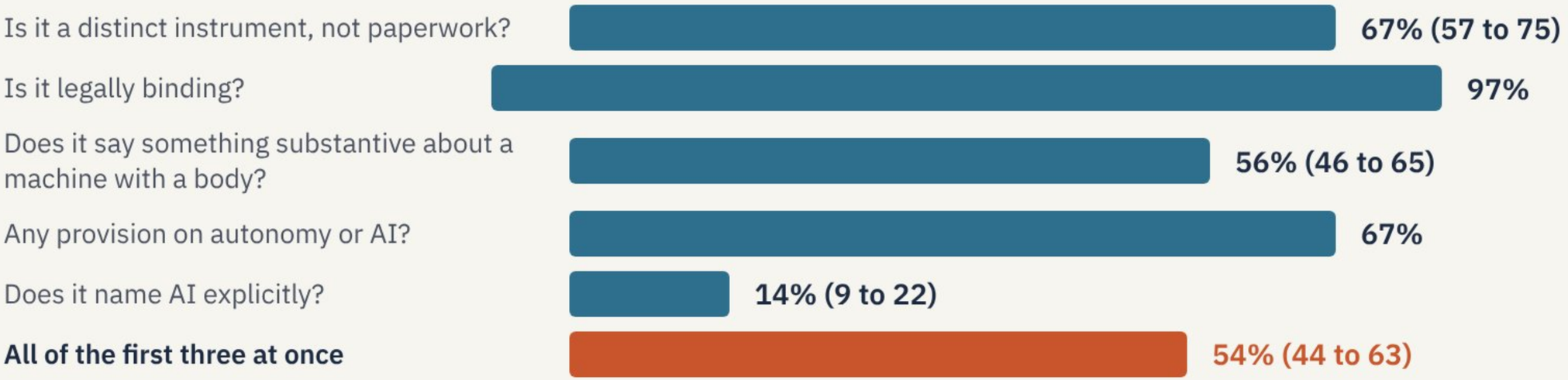
All of these are machine-against-machine. They show the labels are stable, not that an expert would agree with them.

Source: paper, Sections 4 and 5.8, Table 26 and Figure 43; supplement S4 and S9b

THE EVIDENCE · PRECISION

The audit: 100 binding rules, re-read one by one

An independent AI reader took a random sample of 100 of the 299 texts the earlier version had counted as binding robot rules and judged five questions from each text. The ranges are what the true share could plausibly be, given a sample of 100.



So of the 299, about **161** (132 to 190) were genuine. The counting rules that replaced that count keep 195. The two bracket the truth.

Rules and auditor agree on 77 of the 100 (kappa 0.53). The rules keep 15 the auditor rejects and drop 8 the auditor keeps: rules cannot see whether a text has an operative robot clause.

Source: paper, Sections 4 and 5.1, Table 62; supplement S13

THE EVIDENCE · COVERAGE

The coverage audit: 39 rules everyone knows, how many are in

22 of 39
present as a principal instrument whose text was read (the strict test)

31 of 39
present on a looser test that accepts a related text with a matching title

11 of 39
what the first round alone held on the strict test

Still missing on the strict test (17)

Germany's 2021 autonomous-driving law · Korea's Intelligent Robots Act and AI Basic Act · the revised EU Product Liability Directive · Shenzhen's connected-vehicle regulation · Beijing's humanoid-robot measures · UN Regulation 171 · Taiwan's industrial-robot safety standards · the OECD AI Principles as a document · eight safety standards (ISO/TS 15066, IEC 61508, ISO 26262, ISO 21448, ISO/IEC 42001, IEEE 7001, ANSI/RIA R15.06, ANSI/A3 R15.08)

Confirmed to exist, impossible to download

Hangzhou's robot regulations, three Shenzhen special-zone regulations, Hong Kong's autonomous-vehicle regulation, Korea's robot and vehicle decrees, the US Defense Directive 3000.09, California's SB 243, India's Drone Rules 2021, the EU Toy Safety rules, France's 2021 vehicle-liability ordinance and decree, Switzerland's special-category aircraft ordinance, and two standards. Their hosts blocked the search; they head the list for a third round.

Source: paper, Section 5.3, Table 25 and Figure 41; the reference list is data/reference_instruments.csv

READ WITH CARE

How far to trust these numbers

Even 195 is a ceiling

An independent re-reading of 100 randomly chosen texts from the earlier count found 54% to be distinct, binding and genuinely about a machine with a body (plausible range 44% to 63%). About one binding rule in four may still fail a careful reading.

And a floor

Of 39 well-known robot, drone and vehicle instruments, 22 are in the collection with their text. Germany’s autonomous-driving law, Korea’s robot act and eight safety standards are missing; 189 more were confirmed to exist but could not be downloaded.

No person labelled anything

Every label, every collector in round 2, every reviewer and the auditor was an AI model from one company. The checks show the labels are consistent, not that an expert would agree with them.

The collection is lopsided

75% English, 42% European, and the countries with open legal databases (US, EU, Japan) supply 112 of the 195 binding rules. Old statutes carry old dates even when their robot clauses are recent.

The study was reviewed by six adversarial reviewers before this version, and every number was audited against the data. The reviews, the audit and a change log ship with the data, so each claim can be re-examined.

Source: paper, Sections 5.1 and 7; review log

IN PRACTICE

What it means for three people

A worker beside a robot arm

Covered, but by machinery and workplace law, not robot law. ISO 10218 sets the safety requirements for robots working near people; in Europe the Machinery Regulation requires testing and a CE mark before sale. The 13 binding rules that reach factory arms were all written for machines in general.

Someone buying a companion robot for a parent

No binding rule was written for it. What applies is the EU AI Act (bans on manipulative systems; a duty to tell people they are talking to a machine), product-safety and data-protection law, and the voluntary care-robot standard ISO 13482. The gap between research and law is widest here.

A city buying delivery robots

Meets real rules. US states have “personal delivery device” statutes that admit robots to sidewalks under weight, speed and insurance conditions; Japan’s Road Traffic Act treats small remote-operated vehicles as pedestrians once the police are notified. Fifteen binding rules mention delivery robots.

The pattern repeats: the more a robot resembles something already regulated (a vehicle, an aircraft, a machine), the more law it meets. The robots designed to be unlike anything before meet the least.

Source: brief, “What it means for you”; paper, Sections 5.3 and 6

WHAT COMES NEXT

What a third round would do

The paper lists what it could not do. Each item is a known gap with a known fix, in rough order of how much it would change the picture.

1 · READ WHAT IS BLOCKED

Fetch the verified-but-unread instruments from Korea, China's provinces, Hong Kong, France, India and the US Defense Department by other routes. France, Korea, Taiwan and Hong Kong are thin for no reason but access.

2 · HAVE PEOPLE LABEL A SAMPLE

No person has labelled any document. A human-coded sample would turn the consistency checks into a validity check, and allow the counts to be corrected for measurement error.

3 · DATE THE VERSIONS

Old statutes carry old dates even when their robot clauses were added in 2019 to 2023. Recording when each provision arrived would fix the time analyses.

4 · READ THE LONGEST LAWS IN FULL

The model reads up to 110,000 characters; the longest statutes exceed that, and robot articles beyond the cut were sometimes labelled from the model's background knowledge.

5 · RECORD WHOM A RULE BINDS

The codebook does not record whom an instrument binds, when it applies, or the legal effect of a standard (harmonised, incorporated, mandatory). A policy-grade inventory needs all three.

6 · SPLIT THE GAP BY SOURCE

The research-versus-rules gap inherits the second round's search terms, written for drones, vehicles and machinery. Computing it separately by collection source would show how much is real.

Source: paper, Section 7 (Limitations)

OPENNESS

What was released, and what it cost

Everything needed to check a claim ships with the study, so each number traces to a document and each label to the text the model read.

In the release

- The master table: one row per document, every field, the counting unit, families and AI tiers
- Every label with the model, prompt hash, policy, cost and timestamp; the exact model inputs for the main runs
- The codebook, the counting rules, the 157 corrections with reasons, the reference list
- All tables, figures and their data; the dashboard; every script, so the pipeline re-runs from the stored labels
- The six adversarial reviews, the precision audit, the numbers audit and the change log

Not released: the source texts themselves, for copyright reasons. The stored model inputs are the record of what the model saw.

334 USD

total cost of labelling 3,295 documents, re-reading 422, and running six consistency checks

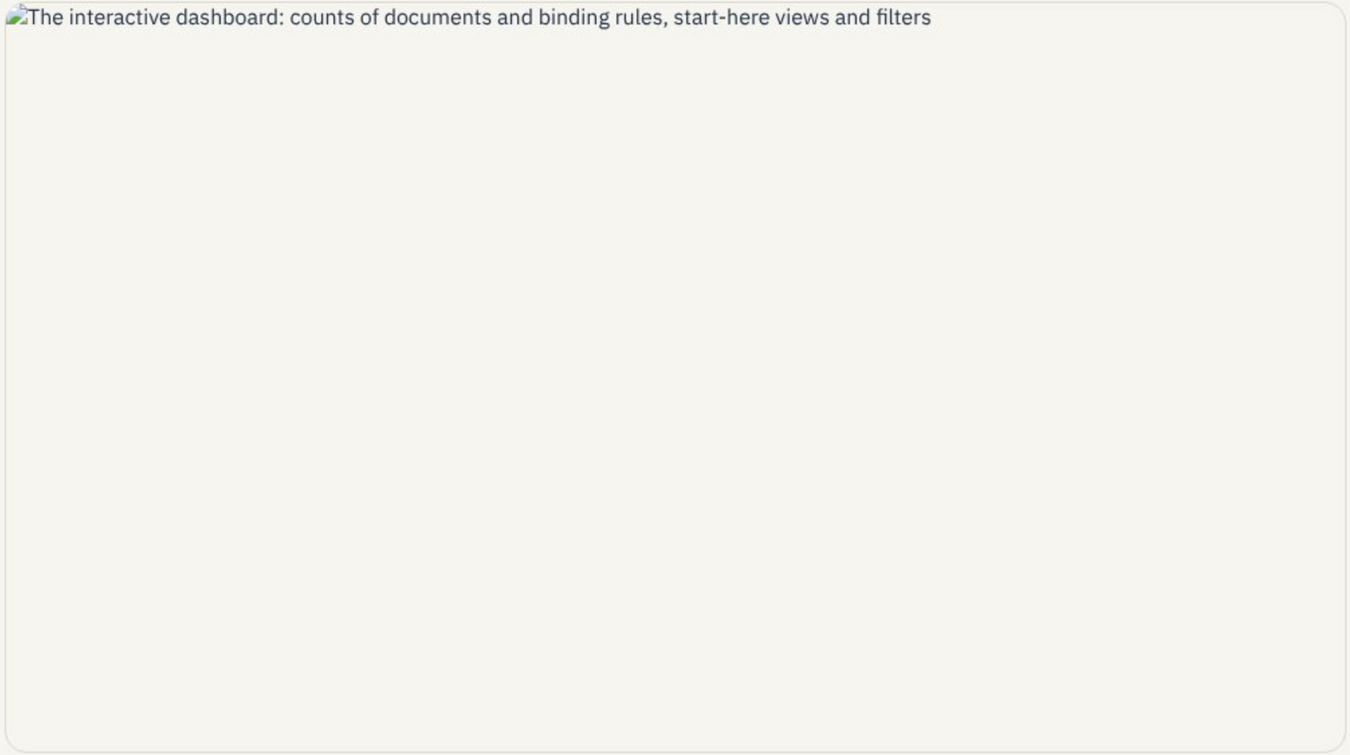
Run	USD
Round 2 labelling (1,720 texts)	126
Round 1 labelling (two runs)	99
Six consistency checks	70
Longer re-read of governance texts	36
Chinese, Japanese and Korean repair	2.5

Caveat: the model's randomness cannot be fixed, so re-runs reproduce the analysis from the stored labels, not the labels themselves.

Source: paper, Section 4 (Annotation) and the Appendix on materials and availability

GO FURTHER

Explore the data yourself



The dashboard

Every document, searchable and filterable by region, machine type, kind of rule and collection round. Click a row to read its summary.

claude.ai/artifact/YKsXJRaT7tSfRXseuo5kuZ

The paper, data and code

The full paper, the analysis report, the data tables, the reviews and the scripts that reproduce every figure.

github.com/cjimmylin/embodied-ai-governance

The dashboard works in a browser without any installation

TO REMEMBER

Five things to remember

1. **Robots are governed sideways.** Of 195 binding rules that reach them, 15 name AI. The rest are aviation, road, machinery, product and medical-device law.

2. **The body is what makes a robot rule different.** Physical safety, testing before sale, how much the machine may decide, and who pays for damage. These hold however the data are sliced.

3. **Research and law look at different robots.** Companion robots are 43% of research papers and 3% of rules, with no binding rule at all. Drones and vehicles are the reverse.

4. **Mention is not detail.** Six binding rules in ten raise the control question; one in four answers it.

5. **Where you look changes what you find.** Five, then 299, then 195 binding rules from the same world. Any count of “AI regulation” should say how it was collected and what it counts.